ABSTRACT

The treatment of infectious diseases faces substantial challenges due to the scarcity of effective antibiotics, leaving many conditions without reliable interventions. This study leverages big data and machine learning (ML) technologies, particularly focusing on drug repurposing (DR), which explores new therapeutic applications for existing drugs. Our goal was to devise an ML-driven DR model that identifies potential associations between an infectious disease and drugs. This study focused on identifying drugs that are repurposable to the treatment of pneumonia due to Streptococcus pneuomiae (ICD-10 code: J13). For our analysis, we sourced drug information (e.g., chemical structure, toxicity, and indication) from DrugBank. Utilizing a pretrained Bidirectional Encoder Representations from Transformer (BERT) model, we derived drug embeddings, which were used to construct a drug-drug similarity network. In parallel, disease embeddings were obtained by using ICD2Vec, leading to the establishment of a disease-disease similarity network. Building upon Cdataset as the benchmark for drug-disease associations, we developed a deep neural network (DNN) that extracts features from the similarity networks and predicts drug-disease associations. Our model exhibited an accuracy exceeding 0.98 and an area under the receiver operating characteristic (AUROC) curve of over 0.83. Notably, our model identified several drugs as candidates for repurposing in the treatment of pneumonia: promazine, fluoxetine, lidocaine, cimetidine, and epirubicin. We successfully developed an ML model that generates repurposing signals for the treatment of pneumonia.

INTRODUCTION

The ever-growing threat of infectious diseases continues to be a serious concern in global health [1]. With the advent of microbial resistance and the decline in the discovery of novel antibiotics, many infectious diseases, which were once considered manageable, are reverting to being life-threatening conditions [1, 2]. This challenge has been further intensified by the laborious and often inefficient process of traditional drug development [3]. As such, the race to find reliable interventions for infectious diseases has never been more urgent.

Pneumonia, a respiratory infection primarily caused by bacteria, viruses, or fungi, is among the leading infectious killers worldwide [4]. In particular, pneumonia due to Streptococcus pneumoniae (ICD-10 code: J13) has posed a significant health challenge [5]. Although vaccines and some antibiotics have historically been effective, emerging antibiotic-resistant strains, coupled with a lack of breakthrough antimicrobial drugs, have left a palpable unmet need in the realm of treatment [6].

Amidst these challenges, drug repurposing (DR) has emerged as a promising strategy [7]. The concept of DR, which involves finding new therapeutic uses for already approved drugs, offers a potentially swift and cost-effective approach [8]. Given that the safety profiles of these drugs are well-documented [9], DR could drastically reduce the time and cost associated with traditional drug discovery, providing a faster route to address the urgent needs of infectious diseases like pneumonia.

In parallel with the rising interest in DR, machine learning (ML) has shown exceptional promise in accelerating drug discovery [10]. By leveraging vast datasets and computational power, ML can sift through numerous drug-disease combinations, identifying associations that might otherwise go unnoticed. The fusion of DR and ML has the potential to revolutionize the way we approach the treatment of infectious diseases [11].

In this context, our research aimed to utilize the synergy between ML and DR, focusing on the identification of potential drugs for the treatment of pneumonia caused by Streptococcus pneumoniae. We set out with a specific objective: to create an ML-driven DR model that efficiently maps the vast landscape of existing drugs and their potential applicability in

combating pneumonia. This approach not only bridges the gap between computational biology and clinical therapeutics but also ushers in a new era of precision medicine in infectious disease management.

METHODS

Drug Data Collection

Comprehensive drug information was sourced from the DrugBank database, a comprehensive online database that offers extensive information about drugs, their molecular targets, and drug actions. Each drug profile in DrugBank includes its chemical structure in various formats, insights into its molecular targets, biological pathways, and potential drug-drug interactions. The database also cross-references its entries with other prominent databases, ensuring a holistic understanding of the drug and its context. We obtained the following drug information: chemical structure in Simplified Molecular-Input Line-Entry System (SMILES), toxicity, indication, and Anatomical Therapeutic Chemical (ATC) classification code. SMILES is a widely-used notation system that provides a way to represent chemical structures as textual strings. This notation makes it convenient to store, search, and share chemical information in databases and software applications without resorting to graphical depictions. For our analysis, chemical structures from DrugBank were represented using their canonical SMILES strings. This standardized representation ensured uniformity and reliability in our data. Likewise, both toxicity and indication information were free-text narratives. In this study, we used the 1st level of ATC code, which was used as the label for the drugs. We extracted the information on both FDA-approved and investigational drugs.

Disease Data Collection

We extracted "Clinical information," "Approximate synonyms," and "Identifiers" from the 10th revision of International Classification of Diseases (ICD-10). ICD-10, developed by the World Health Organization (WHO), is an international standard for diagnosing and classifying health conditions and diseases. Introduced in 1992, it provides a detailed and standardized nomenclature for a wide range of health conditions, aiding in precise diagnosis, epidemiological research, healthcare management, and medical billing. The identifiers (codes) are alphanumeric, ranging from 3 to 7 characters, and categorize diseases based on physiological systems or causative factors, with added granularity to convey clinical specifics. In this study, we used the 1st letter (category) of ICD-10 codes as the label for the diseases.

Drug Embeddings

We first obtained a drug embedding based on SMILES. Here, rather than extracting fingerprints from SMILES, we treated it as a string and applied natural language processing (NLP) technique. To do so, we applied ChemBERTa [12], which generated a 768-dimensional drug embedding. ChemBERTa is a Bidirectional Encoder Representations from Transformer (BERT) model [13] fine-tuned for chemical applications, particularly utilizing chemical tokens from SMILES. Pre-trained on vast chemical datasets and fine-tuned for specific tasks, ChemBERTa offers a bidirectional understanding of chemical context. It has applications in predicting molecular properties, chemical reaction outcomes, molecule classification, and drug discovery, making it a powerful tool for decoding intricate chemical data and accelerating research in the domain.

Likewise, we obtained drug embeddings based on toxicity and indication, separately. For this, we used BERT model to tokenize and embed the free-text descriptions, which resulted in 768-dimensional drug embeddings. Finally, we concatenated the three drug embeddings to obtain the final drug embedding (768 dimensions) to construct drug-drug similarity network.

Disease Embeddings

For diseases, embeddings were obtained by using the ICD2Vec model [14]. This approach is designed to generate continuous vector representations of diseases based on their ICD10 codes. In this study, the focus was primarily on the embeddings associated with the ICD10 code: J13, corresponding to pneumonia due to Streptococcus pneumoniae. We used BERT model to obtain disease embeddings through ICD2Vec workflow, which generated 768 dimensions.

Construction of Similarity Networks

By employing the derived drug embeddings, a similarity network was established. We calculated cosine similarity between the drug vectors in the drug embedding (Equation 1). This network represents the relative similarity between drugs based on their chemical structure, toxicity, and indication. Similarly, with the obtained disease embeddings, a disease-disease similarity network was constructed by calculating cosine similarity between the disease vectors in the disease embedding (Equation 1). The network contains the relationship and similarity between various diseases.

$$1 - \frac{\sum D_i D_j}{\sqrt{\sum \mathrm{D_i^2}} \sqrt{\sum \mathrm{D_j^2}}}$$

Equation 1. The equation for cosine similarity between any two drugs D_i, D_j (or diseases)

Ground-Truth Drug-Disease Association Dataset

Cdataset [15], a benchmark repository for drug-disease associations, was employed. The dataset was published by Luo et al., which is the expansion of a manually curated drug-disease association dataset created by Gottlieb et al. Cdataset contains 2,352 known drug-disease associations between 663 unique drugs and 409 unique diseases. This dataset was used for the training and validation of the deep neural network (DNN) model. In our study, any drug-disease association that was not present in Cdataset was assumed to be negative (i.e., no association).

Construction of Heterogeneous Network and Feature Extraction

We adopted the workflow of Liu et al. [16]. Using the drug similarity network (S_D) and Cdataset, we created a drug feature vector for drug D_i as

$$F_{Di} = concat(M_{ij}, S_{Di}),$$

where M_{ij} is the set of diseases known to be associated with D_i.

Likewise, we constructed a disease feature vector for disease M_i based on disease similarity network (S_M) and Cdataset as

$$F_{Mi} = concat(D_{ij}, S_{Mi}),$$

where D_{ij} is the set of drugs known to be associated with M_i.

Finally, we concatenated drug and disease feature vectors as the final input of the model, which is defined as

$$F_{DiMi} = concat(F_{Di}, F_{Mi})$$

Doing so, we created heterogeneous network-based features that could be less sparse and more informative.

Neural Network Model Development

A deep neural network (DNN) model was developed to predict drug-disease associations. The model integrated features from both the drug-drug and disease-disease similarity networks. The DNN comprised three hidden layers (fully connected layers) with each hidden layer including

dropout layers (Dropout rate = 0.2). The output layer that provided a probability score for potential drug-disease associations. The model was trained using Adam as the optimization algorithm with a binary cross-entropy as the loss function. The learning rate, batch size, and other hyperparameters were optimized based on a validation set. Model performance was primarily gauged using accuracy and the area under the receiver operating characteristic (AUROC) curve.

Drug Repurposing Identification

Upon model development, it was deployed to predict potential drugs that can be repurposed for the treatment of pneumonia due to Streptococcus pneumoniae. Drugs with high probability scores were presented as repurposing candidates.

RESULTS

Drug Embedding

In DrugBank, 10,0003 chemical compounds (including FDA approved drugs and investigational drugs) had SMILES information. Thus, we obtained the drug embedding based on SMILES using ChemBERTa (Figure 1). The drugs were classified according to their 1st level ATC codes. For example, DB00030 (insulin, human), whose ATC code is A10AB01, was classified according to its 1st level ATC code, A.

In total, we had 14 classes of the drugs. In the 2 dimensional t-distributed Stochastic Neighbor Embedding (2D t-SNE) representation of the drugs, we were barely able to observe notable clusters. The lack of notable clusters based on the 1st level ATC codes indicates that SMILES information, or molecular structure, may have weak association with drug uses (as indicated by the ATC codes).

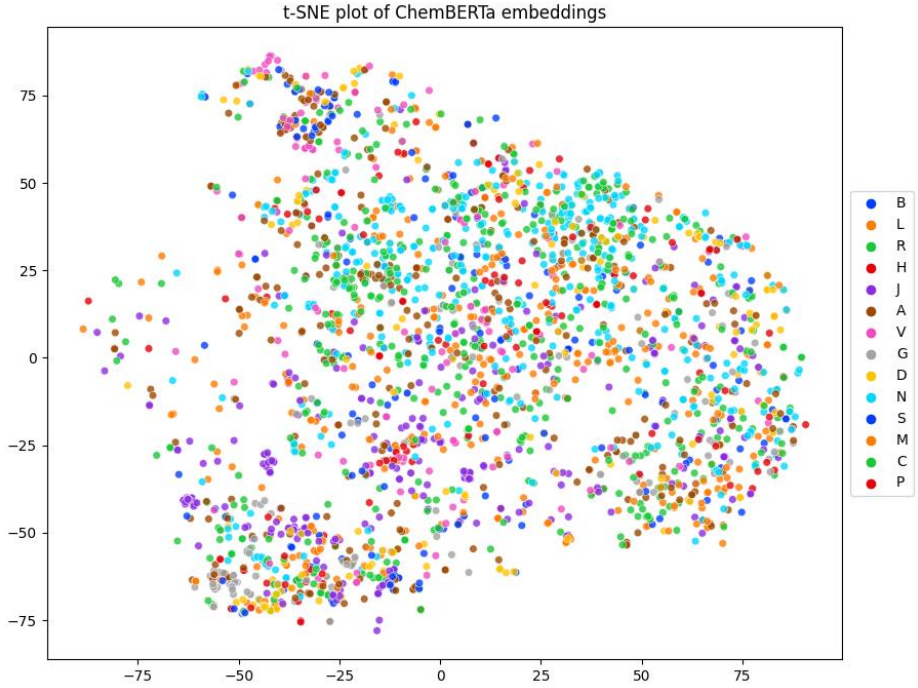


Figure 1 The drug embedding based on SMILES.

In DrugBank, 2,366 and 4,103 drugs had toxicity information, respectively. In order to incorporate toxicity and indication information of drugs, we eliminated those drugs without any of the data. In total, we had 1,907 drugs with SMILES, toxicity, and indication information. After toxicity and indication embeddings were concatenated to the SMILES embedding to create the final embedding, we were able to observe a better cluster in the final embedding

(Figure 2). For example, the 1st level ATC code of C, which includes drugs acting on cardiovascular system, had notable clusters. Moreover, it was observed that after concatenating three embeddings, the size of the representation decreased, indicating that the representation vectors of the drugs have become better clustered and embedded.

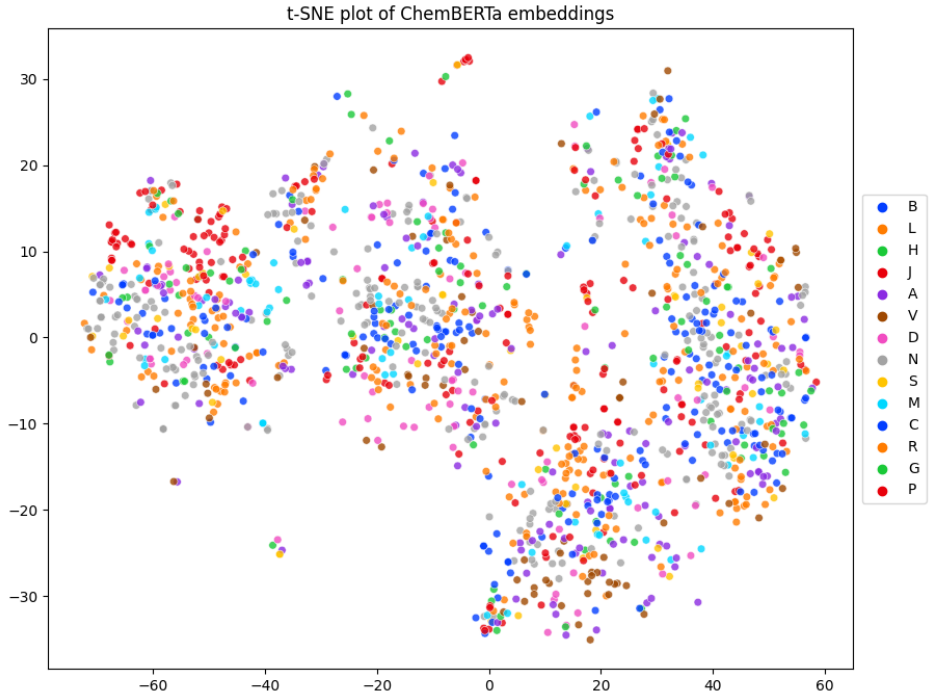


Figure 2 The drug embedding based on SMILES, toxicity, and indication.

Disease Embedding

From ICD-10, we extracted clinical information, synonyms, and identifiers for 43,729 diseases. The ICD-10 codes were transformed so that the diseases are classified according to the first letter of the code. For example, "Cholera, unspecified," whose ICD-10 code is A00.09, was classified according to the first letter of its ICD-10 code, A.

The diseases were embedded based on their clinical information (Figure 3). Notably, diseases of the same first letter of ICD-10 codes were clustered. For example, we were able to observe diseases that belong to ICD-10 code of C, which include malignant neoplasms and benign neoplasms, were situated closely in the representation space. This indicates that the obtained embedding represented the diseases well.

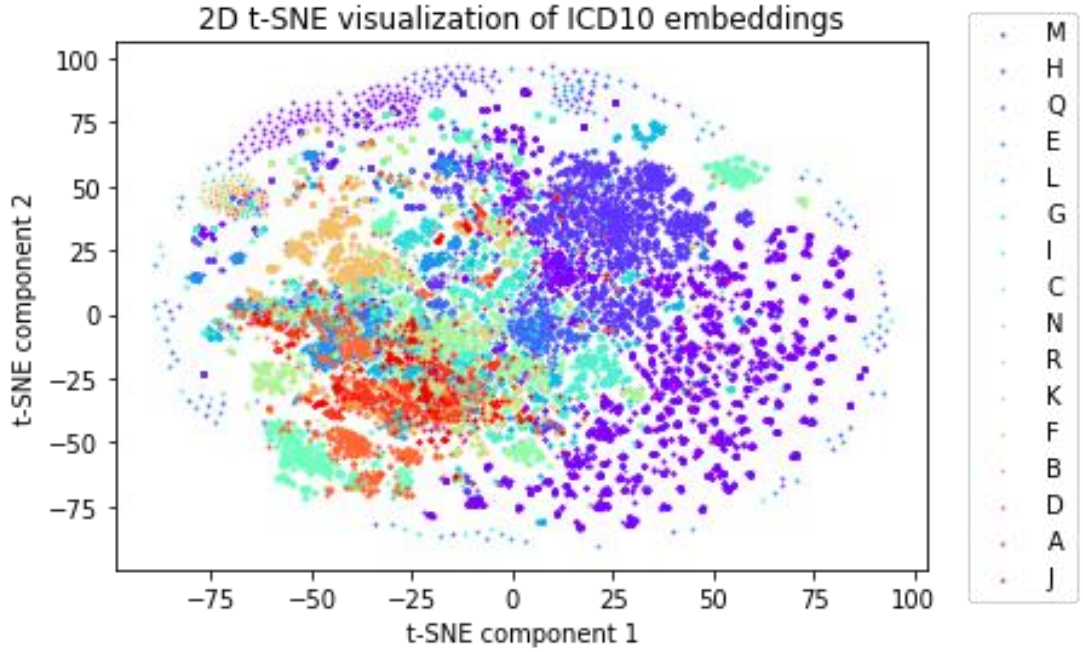


Figure 3 The disease embedding based on ICD2Vec.

Similarity Networks

We calculated cosine similarities between every drug-drug and disease-disease pairs. Because drug embeddings and disease embeddings contain the dense vectors of drugs and diseases, respectively, we assumed that cosine similarity (rather than other methods like Jaccard score) may capture the similarities between drugs (or diseases) better.

The similarity network for 1,907 drugs is shown in figure 4. The similarity network was 1,907 by 1,907 matrix. The diagonal matrix contained identical compounds, indicating cosine similarity value of 1. There was no distinct pattern in the similarities between drugs. Likewise, the similarity network for disease was obtained as a 43,729 by 43,279 matrix, where the diagonal matrix contained the pairs of identical diseases. Similarly, there was no clear pattern in the similarities between diseases.

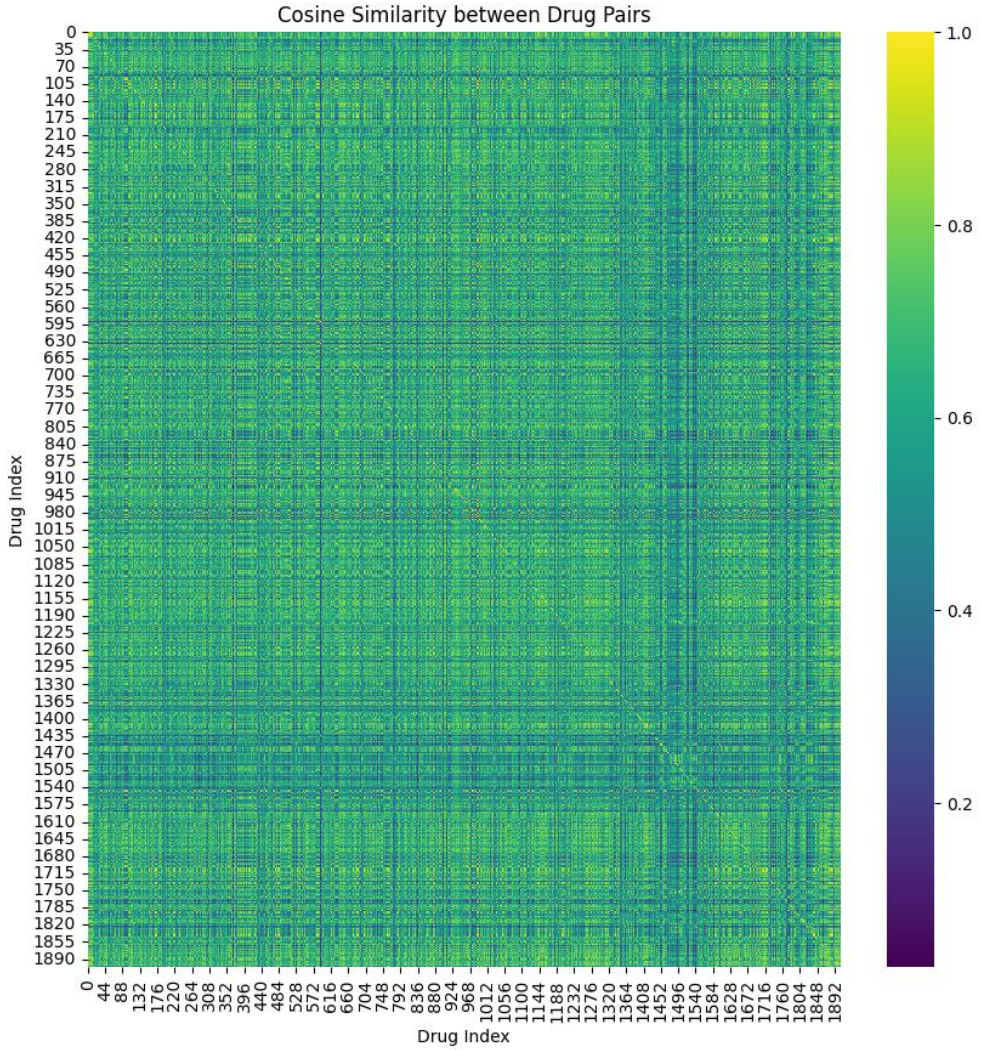


Figure 4 Heatmap for the similarity matrix between 1,907 drugs.

Model Performance

We first extracted features from drug-drug similarity network and disease-disease similarity network to use as input for our DNN model. Using the features as input and the drug-disease association in Cdataset as the label, we defined our DR task as a supervised learning task, where the model was trained to predict an association between a drug and a disease.

The Cdataset contained 2,352 known drug-disease associations and 183,664 unknown (or absent) drug-disease associations. Thus, our model exhibited an accuracy exceeding 0.98. Despite such class imbalance in the dataset, however, our model performed well with the AUROC value of 0.81 (Figure 5). The AUROC value indicates that our DNN model was able to reliably predict the association between drugs and diseases.

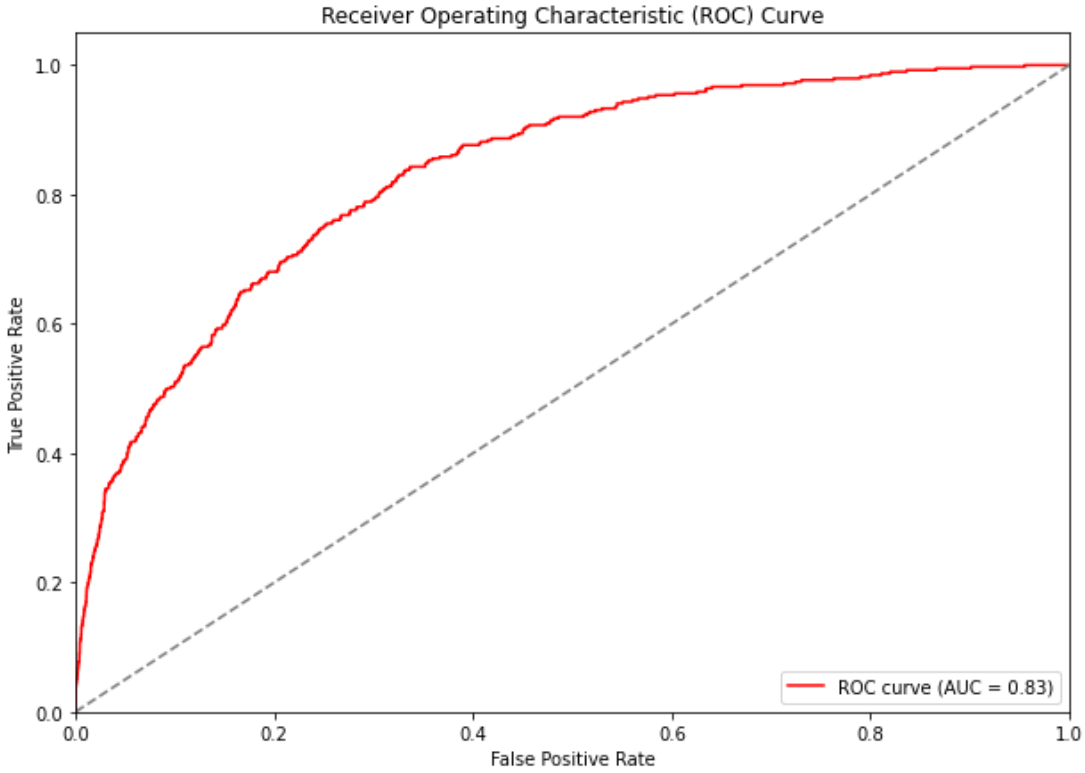


Figure 5 The receiver operator characteristic curve for the drug disease association prediction model.

Drug Repurposing Signals

We defined generating DR signals as predicting drug-disease association, where there was no association in the ground-truth dataset. In other words, if a disease was predicted to be associated with a drug that was not indicated as associated in the Cdataset, we assumed the signal indicates potential possibility of DR.

As a proof-of-concept, we focused on pneumonia due to Streptococcus pneuomiae (ICD-10 code: J13). We used our DNN model to predict five drugs that could be associated with the disease, which did not have a known association according to the Cdataset. The model predicted that promazine, fluoxetine, lidocaine, cimetidine, and epirubicin as potentially repurposable to the treatment of pneumonia due to Streptococcus pneuomiae. Previous studies have reported on the potential antimicrobial activities of the drugs (Table 1).

Table 1 The top 5 drugs predicted to be repurposable to the treatment of pneumonia due to Streptococcus pneuomiae (ICD-10 code: J13).

Drug Name	Current Indication	Supposed Antimicrobial Mode of Action	Tested in vitro on
Promazine	antipsychotics	interaction with bacterial membrane	Staphylococcus aureus, MRSA, Klebsiella pneumoniae
Fluoxetine	antidepressant, Selective serotonin reuptake inhibitor	apoptosis, DNA fragmentation, chromosomal condensation, extracellular phosphatidylserine exposure, membrane damage, efflux pump inhibition, disruption of metabolic processes	S. aureus, MRSA, Staphylococcus epidermis, Enterococcus faecalis, vancomycin-resistant enterococci, Bacillus cereus, M. luteus, E. Coli, A. baumannii, K. pneumoniae, P. aeruginosa
Lidocaine	local anesthetics	disruption of bacterial cell membrane, inhibition of cell wall synthesis, dysfunction of cellular respiration, DNA synthesis alteration, protoplastlysis, altered permeability, leakage of intracellular components, ultrastructural changes, inhibition of membrane-bound enzymatic activities	S. aureus, S. epdermidis, Streptococcus viridans, Streptococcus pyogenes, E. faecalis, E. coli, P. aeruginosa, K. pneumoniae, Neisseria gonorrhoeae
Epirubicin	antineoplastics	interference with cell division	Mycobacterium chelonae, Mycobacterium abscessus
Cimetidine	histamine H2 receptor antagonist	Not known	Not known

DISCUSSION

We successfully developed an ML model that predicts potential drug-disease associations. As a proof-of-concept, we were able to identify drugs with potential antibiotic activities. Of note, our mode demonstrated impressive performance, achieving an accuracy exceeding 0.98 and an AUROC of over 0.83. The results are indicative of a robust and dependable model and underscore the applicability of our approach.

The repurposing of drugs offers a strategic avenue for potentially accelerating the drug development process [17], especially when we consider the challenges of antibiotic-resistant strains and the urgency surrounding infectious diseases like pneumonia. Our machine learning model successfully identified several drugs – promazine, fluoxetine, lidocaine, cimetidine, and epirubicin – as candidates for repurposing in the treatment of pneumonia caused by Streptococcus pneumoniae. Analyzing the mechanisms of action of these drugs can provide insights into their potential efficacy against pneumonia.

Promazine, originally an antipsychotic, acts by antagonizing dopamine D2 receptors [18]. While its neurologic actions are well-documented, promazine, like other phenothiazines, has shown antimicrobial activity [19]. The proposed mechanism involves membrane disruption and ion channel interference in microbial cells [19]. Given that the bacterial cell membrane is crucial for the survival and virulence of Streptococcus pneumoniae, promazine might exhibit direct antibacterial effects or sensitize the bacteria to conventional antibiotics.

Fluoxetine is a well-known selective serotonin reuptake inhibitor (SSRI), primarily used as an antidepressant [20]. Recent studies, however, have highlighted its antimicrobial potential. The exact mechanism remains to be fully elucidated, but some evidence suggests that fluoxetine might interfere with bacterial efflux pumps or disrupt biofilm formation [21]. If validated, this could be particularly useful against persistent Streptococcus pneumoniae infections where biofilms contribute to antibiotic resistance.

Lidocaine, commonly known as a local anesthetic, blocks sodium channels in neuronal cells [22]. Intriguingly, certain bacteria, including Streptococcus pneumoniae, possess sodium channels that are vital for their cellular functions [23]. By inhibiting these channels, lidocaine could exert an antibacterial effect [24]. Moreover, its anti-inflammatory properties might further aid in alleviating the lung inflammation associated with pneumonia [25].

Cimetidine is a histamine H2 receptor antagonist used to reduce stomach acid production in peptic ulcer diseases [26]. Cimetidine has also demonstrated immunomodulatory effects [27]. By augmenting phagocytosis and modulating cytokine release, cimetidine might enhance the host's immune response against Streptococcus pneumoniae. While not directly antimicrobial, it can act in synergy with antibiotics by bolstering the body's defense mechanisms [27].

Lastly, Epirubicin, an antineoplastic agent, interferes with DNA replication in cancer cells [28]. Some antibiotics, like quinolones, target bacterial DNA or its replication machinery [29]. Although a direct antimicrobial activity for epirubicin hasn't been established, its DNA-damaging properties might be harnessed against Streptococcus pneumoniae [30]. However, given its cytotoxic nature, careful consideration is needed before its potential application in infectious diseases.

The drugs identified by our model offer intriguing possibilities for the treatment of pneumonia. Their diverse mechanisms of action hint at multi-pronged strategies—direct antibacterial effects, disruption of bacterial virulence factors, immunomodulation, and possibly synergistic actions with existing antibiotics. Future in vitro and in vivo studies are essential to validate these hypotheses and determine the clinical viability of these drug candidates in the battle against pneumonia.

Despite the results, our study has several limitations. First, while DrugBank is a

comprehensive database, it may not encompass every drug or the most recent data available. Any omissions or outdated information could impact the predictions made by the model. Second, our model was specifically optimized for pneumonia due to Streptococcus pneumoniae. While the approach is generalizable, the performance metrics might differ when applied to other infectious diseases or different causative agents of pneumonia. Moreover, with an accuracy of over 0.98, there might be concerns about overfitting, where the model performs exceptionally well on the training data but might not generalize as effectively to new, unseen data. Finally, our study is primarily computational. The true efficacy and safety of repurposed drugs can only be confirmed through rigorous in vitro and in vivo experimental validations.

CONCLUSION

Today, when antibiotic resistance and drug shortages pose significant challenges, there is a dire need for innovative approaches to identify effective treatments for infectious diseases. This study capitalized on the potential of big data and machine learning technologies to apply the concept of drug repurposing. Our novel machine learning-driven drug repurposing model, which strategically employs a deep neural network to analyze drug-disease associations from datasets like DrugBank and ICD-10, demonstrated an AUROC value surpassing 0.83. By focusing on Streptococcus pneumoniae-induced pneumonia (ICD10 code: J13) as a case study, we've identified several drugs, including promazine, fluoxetine, lidocaine, cimetidine, and epirubicin, as potential candidates for repurposing. These findings not only open doors for further in-vitro and in-vivo validations but also highlight the immense potential of machine learning in facilitating drug repurposing studies. As the global health community grapples with evolving challenges, leveraging technology, data, and computational methods, as showcased in our research, can be pivotal in shaping future medical interventions and ensuring effective treatments for numerous conditions.

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